

1. Sales Order Entry

- Customer order is logged into ERP (via email, portal, or manual entry).
- ERP validates pricing, delivery dates, and customer credit.

2. Manufacturing Order Creation

- ERP automatically generates a manufacturing order linked to the sales order.
- Defines product, quantity, and delivery timeline.

3. Material Requirements Planning (MRP)

- ERP checks raw material inventory.
- Calculates shortages and triggers **purchase requisitions**.
- Links to **Bill of Materials (BOM)** for accuracy.

4. Procurement & Inventory

- Purchase orders sent to suppliers.
- Goods received, inspected, and updated in ERP.
- Materials allocated to production floor.

5. Production Scheduling

- ERP schedules jobs based on machine/work-center availability.
- Production manager notified with real-time updates.

6. Shop Floor Execution

- Work orders dispatched to operators.
- Labor hours and machine usage tracked digitally.
- ERP updates progress and WIP (Work-in-Progress).

7. Quality Control

- QC module ensures raw materials and finished goods meet standards.
- Non-conformance flagged for corrective action.

8. Inventory & Dispatch

- Finished goods moved to warehouse.
- ERP updates stock levels and costing.
- Dispatch scheduled with invoice generation.

Example ERP Flow (Simplified)

Step	ERP Action	Outcome
Sales Order	Order entry	Customer demand captured
MRP Run	Material check	Shortages flagged
Procurement	Purchase requisition	Raw materials secured
Production Scheduling	Job allocation	Optimized timeline
Execution	Work order dispatch	Real-time tracking
Quality Control	QC checks	Standards ensured
Dispatch	Invoice & delivery	Customer fulfillment